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A Comparative Study of Tree-based Models for Healthcare Risk Prediction: Cross-validation and SHAP-based Stability Analysis

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II. Introduction and Research Questions

II.1. Research Context

Healthcare risk prediction has become an important application of machine learning, particularly with the increasing availability of structured healthcare data. Machine learning models can be used to identify patterns in demographic, clinical, and physiological features and estimate the risk of developing a disease or experiencing an adverse health outcome. Such predictive models may provide useful decision-support information for healthcare-related analysis.

Among machine learning approaches for structured data, tree-based models are widely used because they can capture nonlinear relationships and feature interactions while requiring relatively few assumptions about the underlying data distribution. In particular, boosting-based models such as AdaBoost, XGBoost, and LightGBM have demonstrated strong predictive capabilities and are commonly applied to classification problems.

However, healthcare datasets can differ substantially in terms of sample size, feature distributions, missing values, and class imbalance. As a result, the performance of a particular model may vary across datasets. Evaluating multiple models across multiple healthcare datasets under a consistent experimental protocol can therefore provide a more comprehensive understanding of their behavior.

II.2. Research Problem

Although tree-based models can achieve strong predictive performance, comparing models based solely on a single train-test split or a single performance metric may provide an incomplete view of their reliability. A model that performs well on one particular split may exhibit considerable variation when trained on different subsets of the data.

Cross-validation provides a systematic approach for evaluating this variability by repeatedly training and evaluating models on different data partitions. Therefore, model comparison should consider not only average predictive performance but also the consistency of performance across cross-validation folds.

Furthermore, predictive performance does not indicate why a model produces its predictions. In healthcare-related machine learning, identifying influential features can provide additional insight into model behavior. SHapley Additive exPlanations (SHAP) can be used to quantify the contribution of individual features to model predictions. However, feature importance obtained from a single training run may not necessarily remain consistent when the training data changes.

This raises a second stability concern: whether the explanations produced by a model remain consistent across different cross-validation folds. A feature that appears highly important in one fold but substantially decreases in importance in other folds may indicate that the corresponding interpretation is sensitive to the training data.

Therefore, this study considers both predictive stability and explanation stability when comparing tree-based models for healthcare risk prediction.

II.3. Research Questions

Based on the research problem described above, this study investigates the following research questions:

- **RQ1:** How do AdaBoost, XGBoost, and LightGBM compare in healthcare risk prediction across multiple datasets?
- **RQ2:** How stable is the predictive performance of these models across cross-validation folds?
- **RQ3:** How stable are SHAP-based feature importance rankings across cross-validation folds?

II.4. Contributions

This study makes the following contributions:

- **Systematic comparison of boosting-based tree models.** We provide a controlled empirical comparison of AdaBoost, XGBoost, and LightGBM for healthcare risk prediction across multiple datasets using a consistent experimental protocol.
- **Evaluation of predictive stability.** In addition to reporting predictive performance, we analyze the variability of model performance across cross-validation folds to assess the consistency of the models.
- **Analysis of SHAP explanation stability.** We investigate the stability of SHAP-based feature importance rankings across cross-validation folds, providing an empirical assessment of whether model explanations remain consistent under different training subsets.
- **Multi-dataset evaluation.** By evaluating the models across multiple healthcare datasets, the study examines whether observed performance and explanation patterns are consistent across different data characteristics.
- **Reproducible experimental framework.** The experiments are conducted using a standardized and reproducible pipeline, enabling consistent comparison across models, datasets, and cross-validation folds.

III. Related Work

IV. Methodology and Research Architecture

V. Data and Experimental Design

VI. Results and Discussion

VI.1. Summary of Results and Ablation Study

VI.2. Discussion

VII. Conclusion and Future Work

VII.1. Conclusion

VIII. References

[1] M. Mancer et al., “A systematic literature review of crop recommendation systems for agriculture 4.0”, 2025.

[2] R. K. Nde et al., “Crop selection: A survey on factors and techniques”, *Smart Agricultural Technology*, vol. 9, p. 100 602, 2024, ISSN: 2772-3755. DOI: https://doi.org/10.1016/j.atech.2024.100602 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772375524002077

Appendix

1. **AIVIETNAM:** Further information about AIVIETNAM can be found here.
2. **GitHub Repository:**
3. **Dataset:**
4. **Demo Video / Presentation:**
    - Link demo video: Link demo
    - Link slide presentation: Link slide presentation